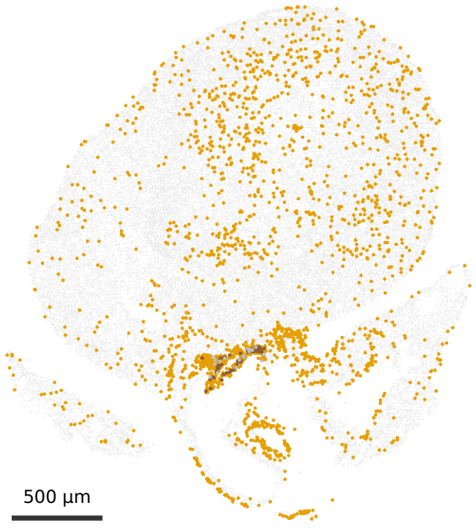
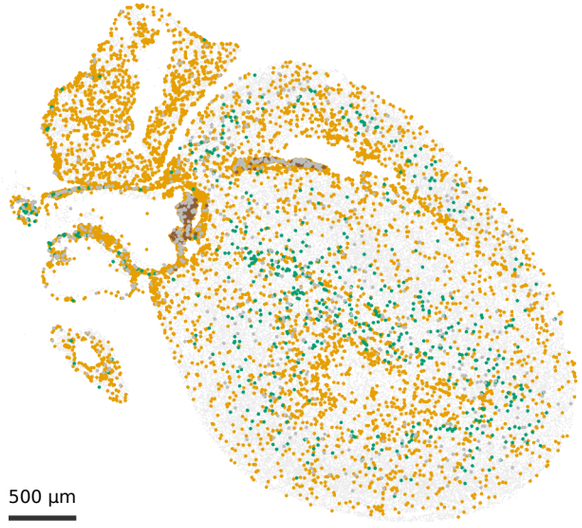
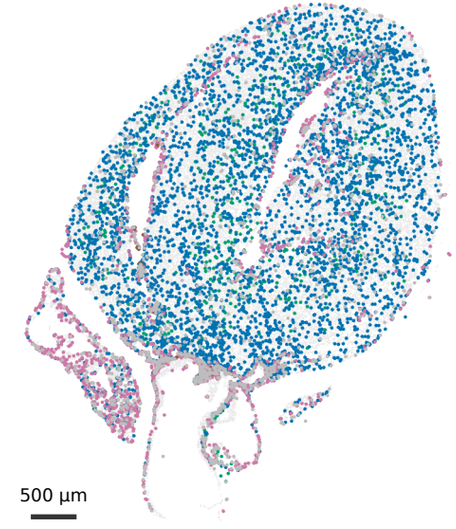
a P0 fibroblast candidates



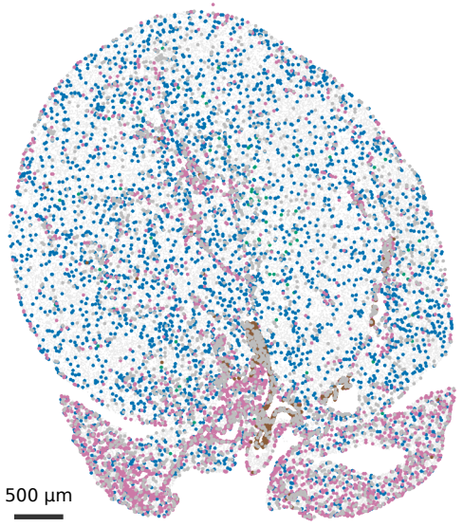
b P7 fibroblast candidates



c P14 fibroblast candidates

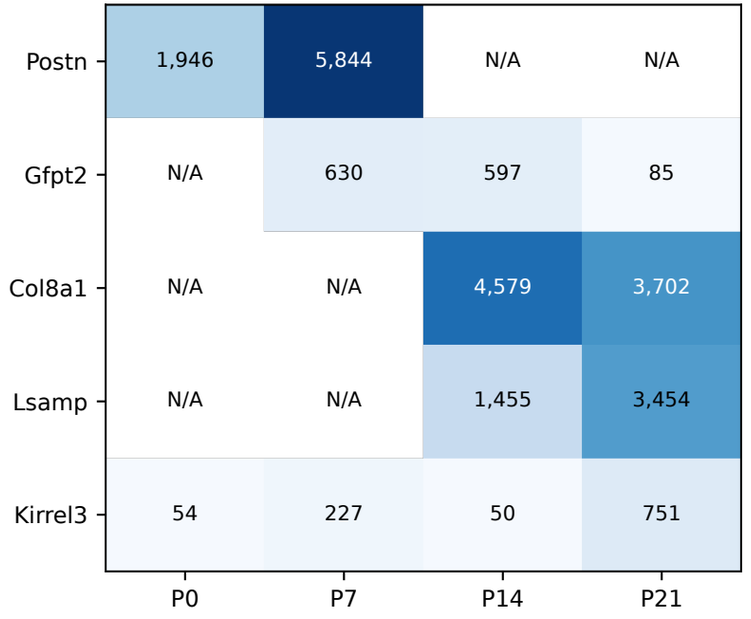


d P21 fibroblast candidates



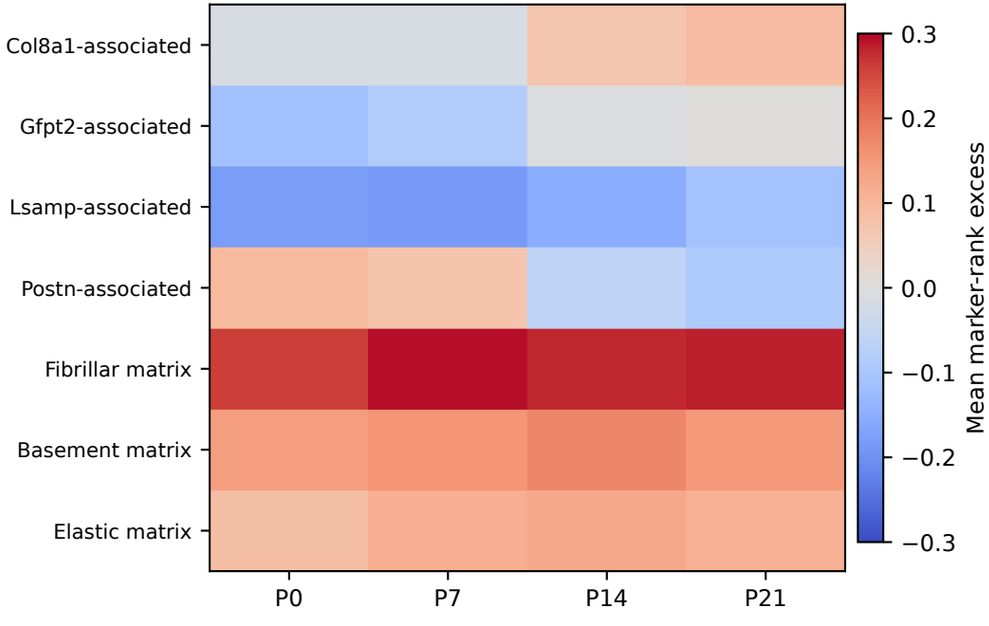
Postn Col8a1 Kirrel3
Gfpt2 Lsamp Unresolved

e Assignments require an available reference



N/A: family not assessable with that age's reference
Counts show coverage, not tissue abundance.

f Gene patterns can be measured at every age

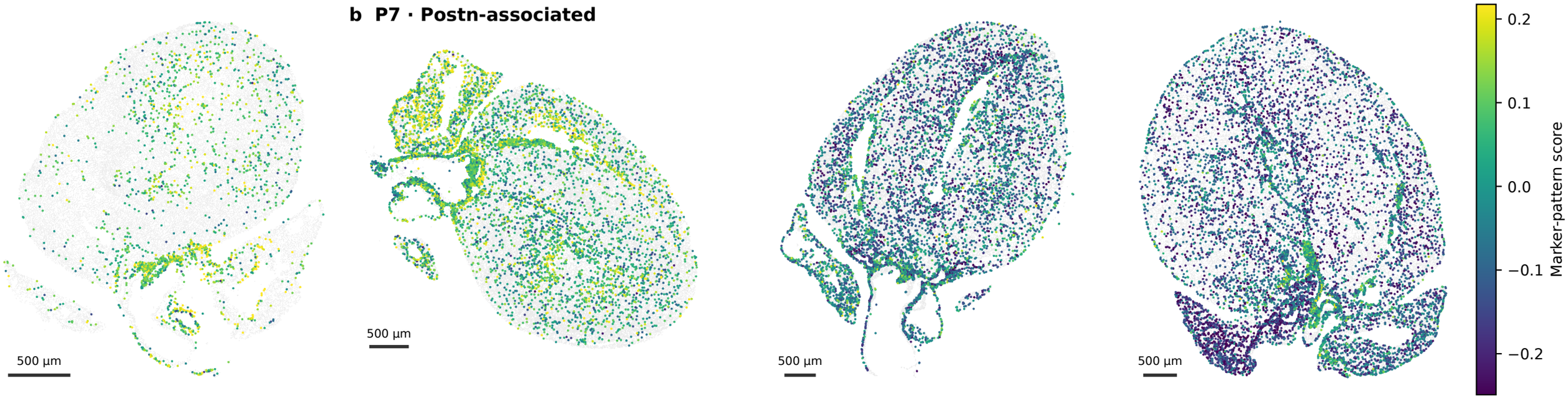


a P0 · Postn-associated

b P7 · Postn-associated

c P14 · Postn-associated

d P21 · Postn-associated

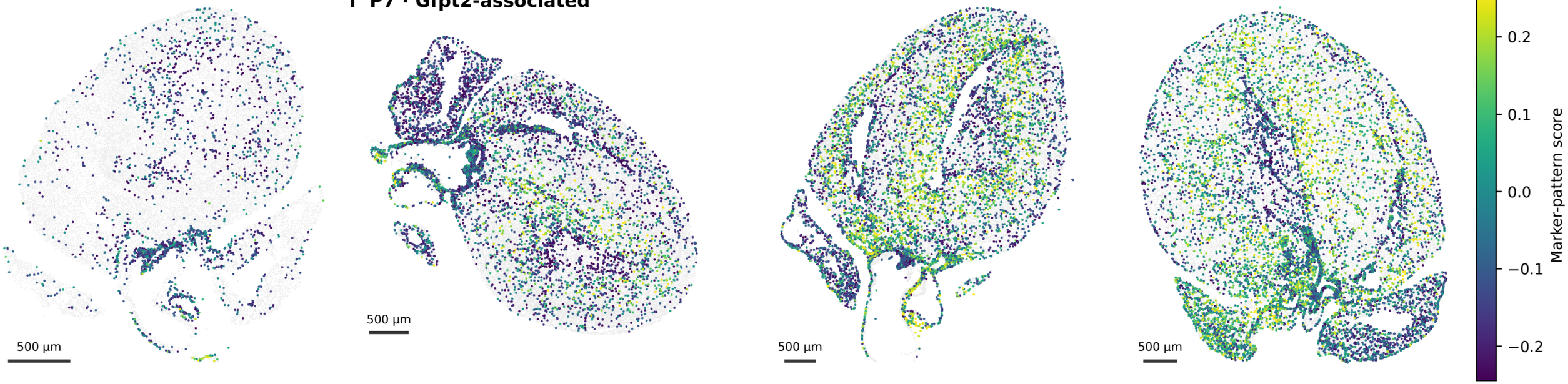


e P0 · Gfpt2-associated

f P7 · Gfpt2-associated

g P14 · Gfpt2-associated

h P21 · Gfpt2-associated

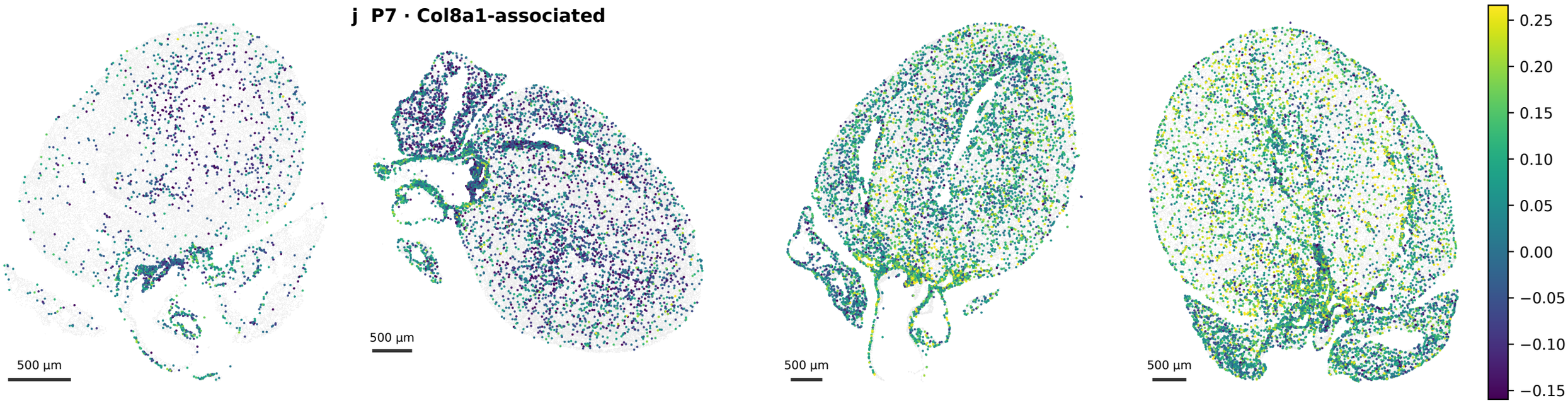


i P0 · Col8a1-associated

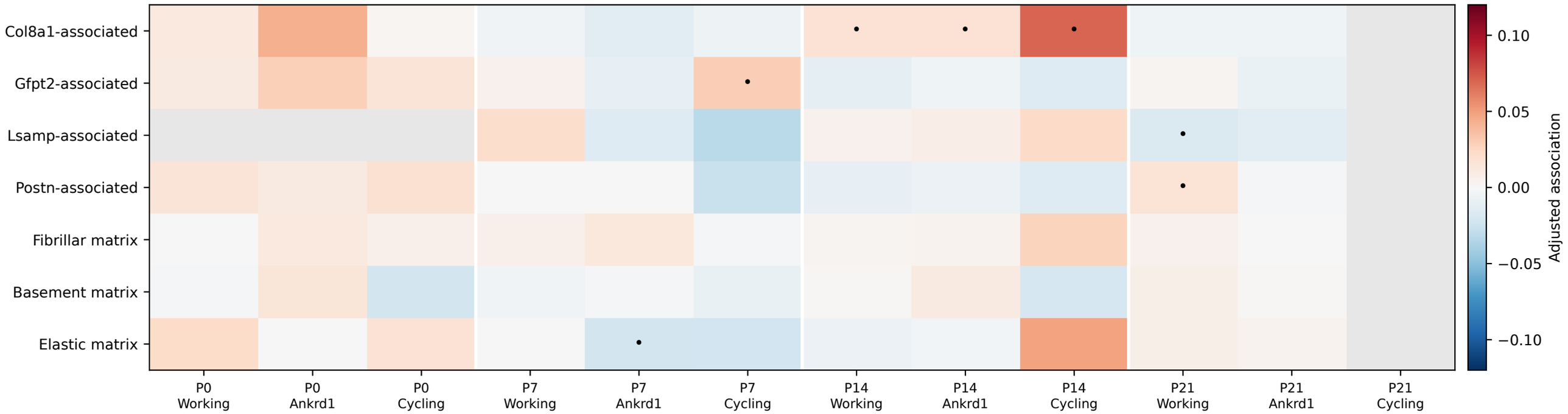
j P7 · Col8a1-associated

k P14 · Col8a1-associated

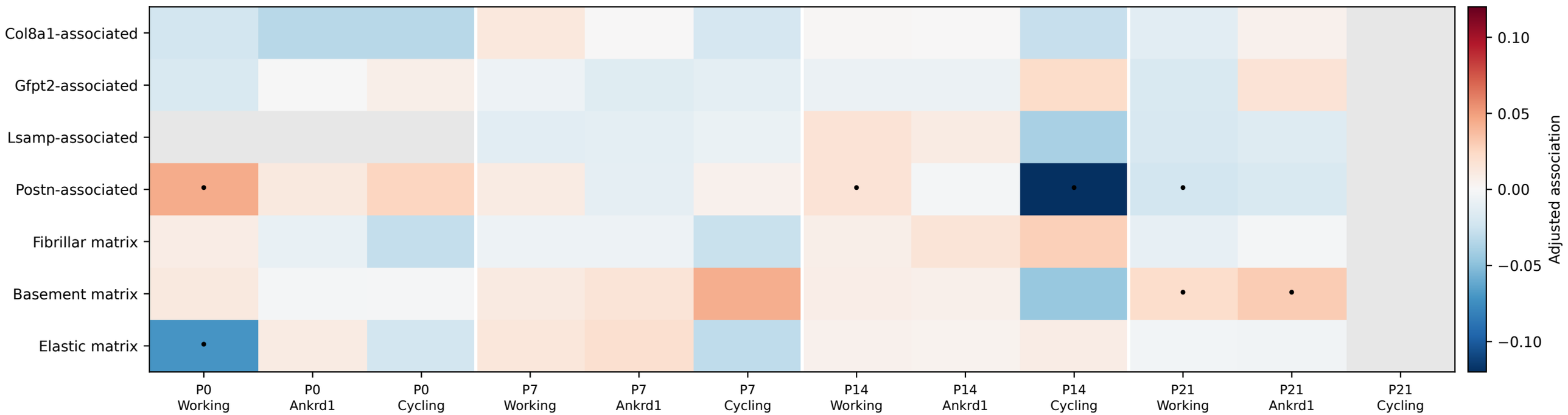
l P21 · Col8a1-associated



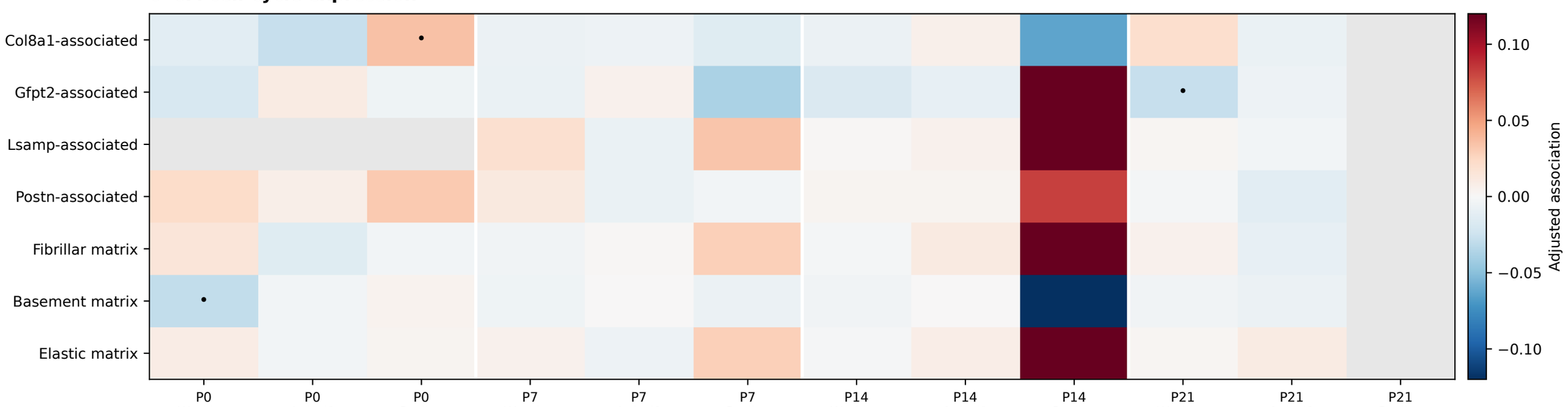
a CM energy-metabolism expression

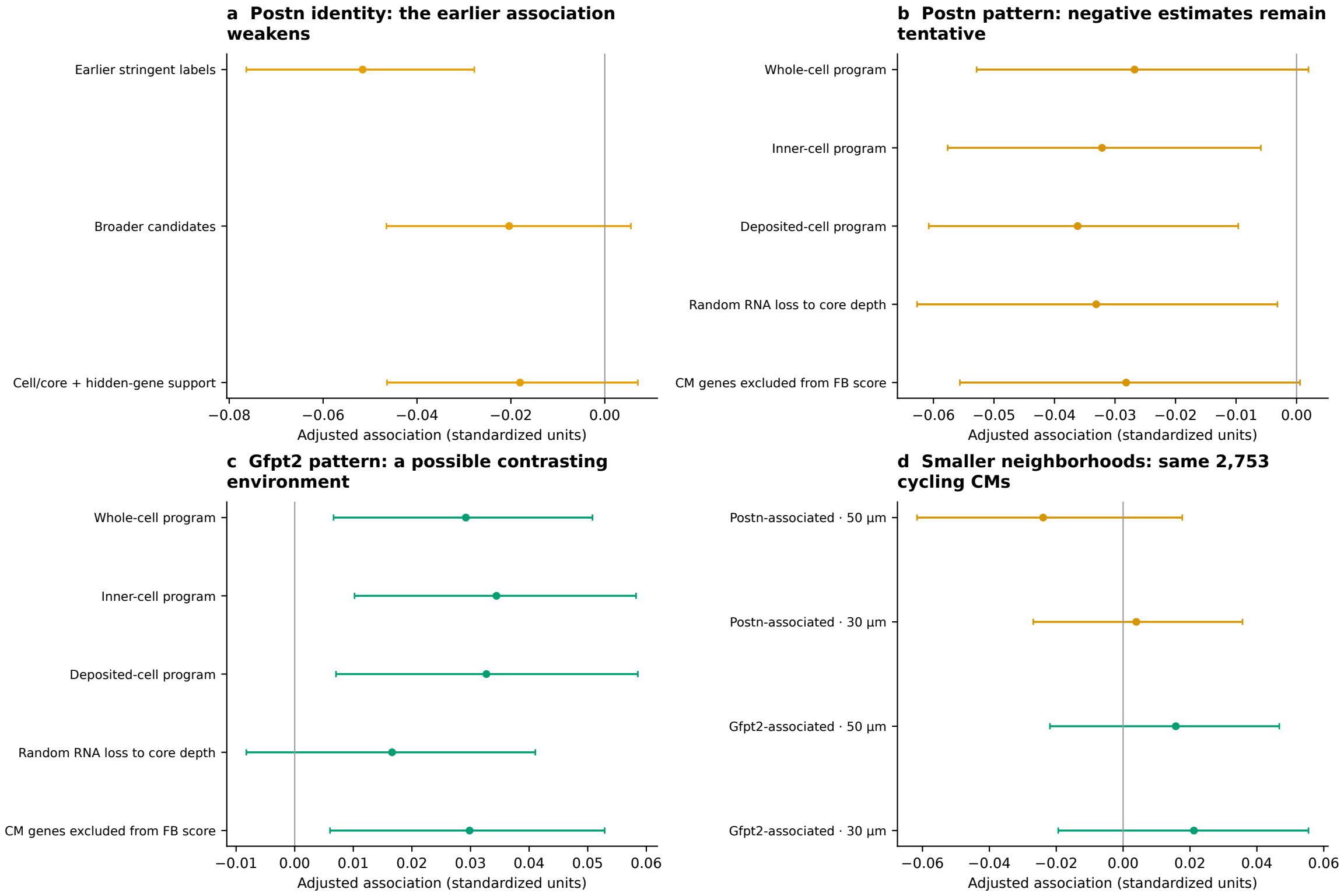


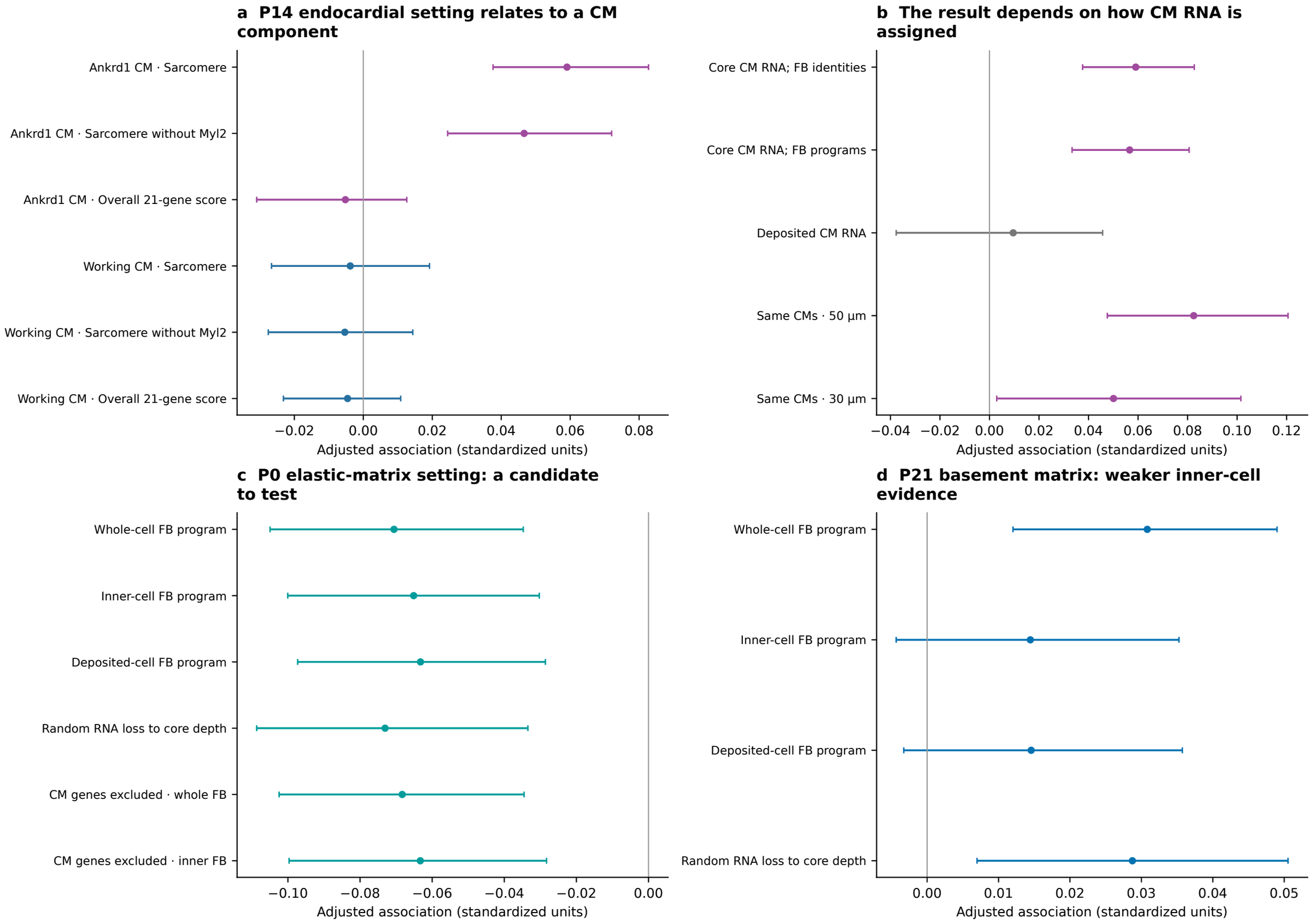
b CM sarcomere expression

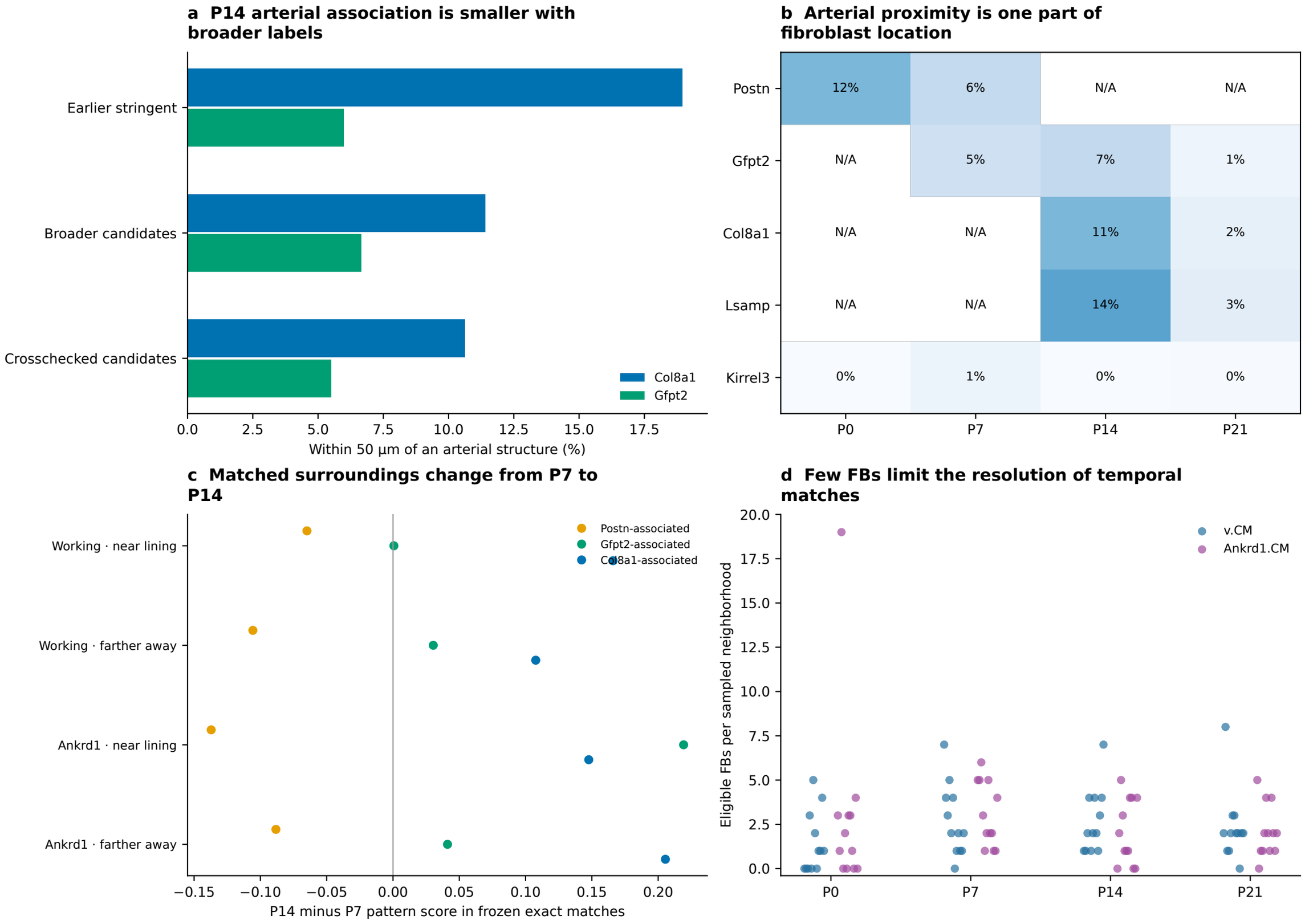


c CM cell-cycle expression

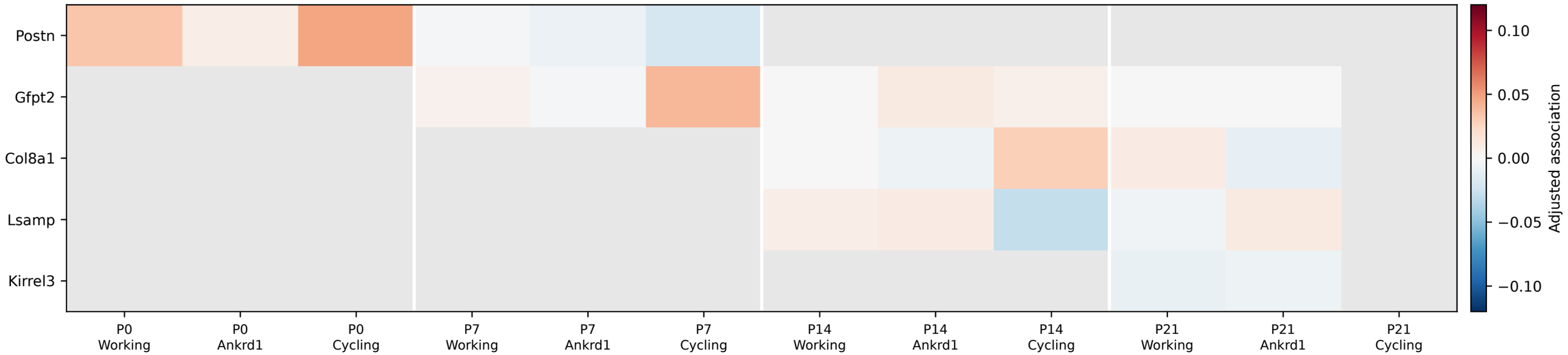



a Postn identity: the earlier association weakens
Earlier stringent labels
Broader candidates
Cell/core + hidden-gene support
−0.08
−0.06
−0.04
−0.02
0.00
Adjusted association (standardized units)
b Postn pattern: negative estimates remain tentative
Whole-cell program
Inner-cell program
Deposited-cell program
Random RNA loss to core depth
CM genes excluded from FB score
−0.06
−0.05
−0.04
−0.03
−0.02
−0.01
0.00
Adjusted association (standardized units)
c Gfpt2 pattern: a possible contrasting environment
Whole-cell program
Inner-cell program
Deposited-cell program
Random RNA loss to core depth
CM genes excluded from FB score
−0.01
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Adjusted association (standardized units)
d Smaller neighborhoods: same 2,753 cycling CMs
Postn-associated · 50 µm
Postn-associated · 30 µm
Gfpt2-associated · 50 µm
Gfpt2-associated · 30 µm
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
Adjusted association (standardized units)

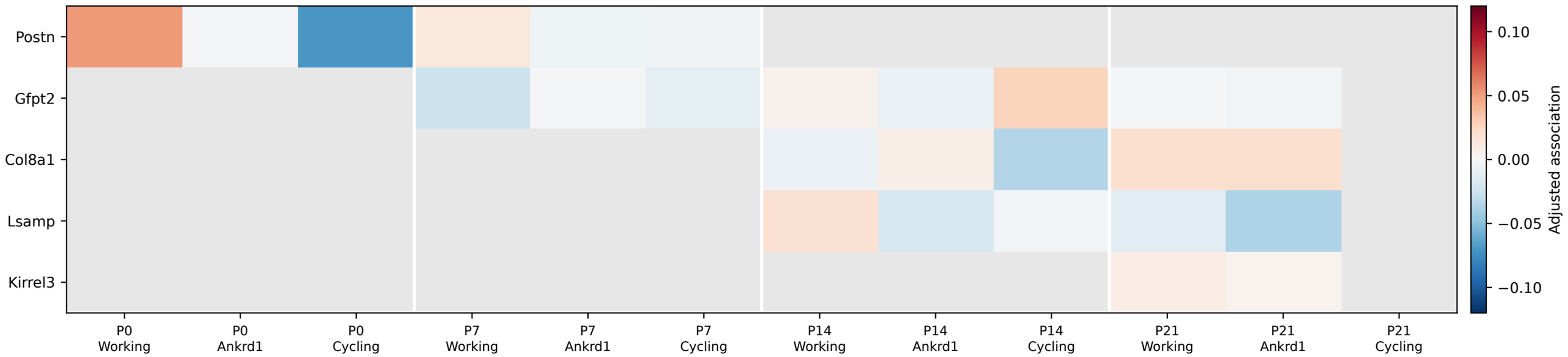

a P14 endocardial setting relates to a CM component
Ankrd1 CM · Sarcomere
Ankrd1 CM · Sarcomere without Myl2
Ankrd1 CM · Overall 21-gene score
Working CM · Sarcomere
Working CM · Sarcomere without Myl2
Working CM · Overall 21-gene score
−0.02
0.00
0.02
0.04
0.06
0.08
Adjusted association (standardized units)
b The result depends on how CM RNA is assigned
Core CM RNA; FB identities
Core CM RNA; FB programs
Deposited CM RNA
Same CMs · 50 µm
Same CMs · 30 µm
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Adjusted association (standardized units)
c P0 elastic-matrix setting: a candidate to test
Whole-cell FB program
Inner-cell FB program
Deposited-cell FB program
Random RNA loss to core depth
CM genes excluded · whole FB
CM genes excluded · inner FB
−0.10
−0.08
−0.06
−0.04
−0.02
0.00
Adjusted association (standardized units)
d P21 basement matrix: weaker inner-cell evidence
Whole-cell FB program
Inner-cell FB program
Deposited-cell FB program
Random RNA loss to core depth
0.00
0.01
0.02
0.03
0.04
0.05
Adjusted association (standardized units)


a P14 arterial association is smaller with broader labels
Earlier stringent
Broader candidates
Crosschecked candidates
Col8a1
Gfpt2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Within 50 µm of an arterial structure (%)
b Arterial proximity is one part of fibroblast location
Postn
Gfpt2
Col8a1
Lsamp
Kirrel3
12%
6%
N/A
N/A
N/A
5%
7%
1%
N/A
N/A
11%
2%
N/A
N/A
14%
3%
0%
1%
0%
0%
P0
P7
P14
P21
c Matched surroundings change from P7 to P14
Working · near lining
Working · farther away
Ankrd1 · near lining
Ankrd1 · farther away
Postn-associated
Gfpt2-associated
Col8a1-associated
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
P14 minus P7 pattern score in frozen exact matches
d Few FBs limit the resolution of temporal matches
Eligible FBs per sampled neighborhood
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
v.CM
Ankrd1.CM
P0
P7
P14
P21

a Metabolic expression



b Sarcomere expression



c Cell-cycle expression

